

AWS CodePipeline

02-mlops-pipelines/codepipeline

Overview

CodePipeline is AWS's managed continuous delivery orchestrator. It connects source, build, test, approval, and deployment actions into a stage based release flow.

Its main value is disciplined automation. Instead of relying on a person to remember deployment steps, the pipeline executes the same controlled sequence every time.

Core Concepts

- A pipeline is composed of stages, and each stage contains actions such as source retrieval, CodeBuild execution, manual approval, or deployment integration.
- Artifacts move from one stage to the next so that the exact built output is what gets tested and deployed.
- Pipelines integrate well with IAM, KMS, EventBridge, and downstream deployment targets such as ECS, Lambda, CloudFormation, or custom actions.

How It Works

- A code or configuration change triggers the pipeline, which pulls the source and passes it into automated build and validation steps.
- Successful outputs are packaged as artifacts, optionally gated by manual approval, then deployed to the target environment through a controlled action.
- Notifications and audit trails make it clear which version moved where and whether a failure happened in source, build, test, or deployment.

AI and ML Relevance

- In ML systems, CodePipeline is often used for infrastructure as code, deployment of inference services, packaging of training images, or promotion of approved model packages.
- It is particularly useful when the broader platform standardizes on AWS native CI and CD rather than an external orchestration tool.

Operational Notes

- Keep pipeline stages purposeful. Mixing unrelated validation and deployment logic into one opaque step makes failures harder to debug.
- Store artifacts securely and make the pipeline idempotent so reruns do not create confusing partial state.
- Use manual approval only where human judgment adds real value. Too many approval steps turn automation into ceremony.

What To Remember

- CodePipeline is a release conveyor belt. It is most valuable when release steps are standardized and auditable.

- Artifacts matter because they preserve what was actually built and tested.
- It solves software promotion, not experiment tracking. That is an important boundary in ML systems.